Circumference of a circle

1

a 50.27 cm **b** 25.13 cm **c** 31.42 cm **d** 43.98 cm

2 169.6 cm

3 119.4 cm

4 2.23 cm

Arc length

5

a 12.6 units **b** 1.8 units **c** 6.2 m **d** 35.4 units

e 62.8 cm

6 $r = 19.76$ cm

7 1.29 m

8 25 408.5 km

Area of circle

9

a 201.1 cm^2 **b** 78.5 cm^2 **c** 153.9 cm^2 **d** 113.1 cm^2

10

a $r = 10.59$ cm **b** 66.5 cm

Sectors

11

a 18.9 cm **b** 18.8 cm

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12

a

i $473.89\ \text{mm}$ ii $11\,744.12\ \text{mm}^2$

b

i $218.75\ \text{cm}$ ii $2683.95\ \text{cm}^2$

c

i $92.40\ \text{m}$ ii $497.59\ \text{m}^2$

d

i $108.91\ \text{mm}$ ii $730.62\ \text{mm}^2$

13

a $P = 257.73\ \text{m}$

b $A = 3112.56\ \text{m}^2$

14

a $P = 586.78\ \text{cm}$

b $A = 15\,925.02\ \text{cm}^2$

Segment

15 $6\pi - 9\sqrt{3}$ square units

16 $16\sqrt{3} + \dfrac{160\pi}{3}$ square units

17

a $\theta = 30^\circ$

b $108\pi - 324\ \text{cm}^2$

c $1188\pi + 324\ \text{cm}^2$

d $11\pi + 3 : \pi - 3$

18

a 73.7°

b $6.4\ \text{cm}$

c $4.1\ \text{cm}^2$

19 $50.7\ \text{cm}^2$

20 $\dfrac{13357\pi}{360} - \dfrac{361}{2}\sin 37^\circ\ \text{mm}^2$

21

a $\theta = \dfrac{240^\circ}{\pi}$

b $14.6\ \text{cm}^2$

Applications

22

a $\theta = 60^\circ$

b $\dfrac{10\pi + 3\sqrt{3}}{2\pi - 3\sqrt{3}}$

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23

a $x = 6$ b $\theta = 62^\circ$ c 9.8 cm^2

24

a $r = 12 \text{ cm}$ b $4\pi \text{ cm}$

25 22.1 laps

26

a 242.9 cm^2 b 242.9 cm^2 c 113.1 cm^2 d 146.1 cm^2

e 261.8 cm^2 f 40.8 cm^2 g 83.9 cm^2 h 176.5 units^2

i 19.6 cm^2 j 226.2 cm^2

27

a 60.68 cm b 43.98 cm

c 42.85 m d 35.00 cm

e 80.40 cm f 31.42 cm

28

a 157.8 cm^2 b 9.4 cm^2

29

a 37.4 cm b 35.7 mm

30 7.2 m

31

a 71 mm^2 b \$284

32 14.1 cm^2

33

a $r = 40.2 \text{ cm}^2$ b $r = 4.9 \text{ cm}$

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